

# Topology of arising and the principle of minimal action in admissibility structures

Denis Britov  
Independent Researcher

## Abstract

This paper develops a Level-0 ontological reconstruction of the minimal structural conditions under which time, space, energy, and stable localized form become admissible. The analysis does not begin from presupposed spacetime, particles, or established physical laws. Instead, it starts from distinguishability within a proto-field of admissibility and asks which structural conditions must hold for directional differentiation, compensation, oscillatory stability, and localized closure to become possible. The mathematical expressions introduced in the paper are used as admissibility formalisms and structural proxies. They are not presented as empirically established microphysical laws, but as compact representations of the conditions under which specific configurations become structurally admissible.

The reconstruction proceeds through four connected moments. First, distinguishability generates temporal and spatial differentiation together with compensation energy as a structural measure of admissible deviation. Second, coherent oscillatory configurations become selectable through a variational admissibility filter, where minimal action functions as a criterion of structural stability rather than as a ready-made law of nature. Third, triadic resonant closure produces the first nontrivial self-supporting contour, introducing internal irreducibility, configurational density, and the conditions for a localized CRT-node. Fourth, interacting localized nodes form ensembles whose collective stability completes Level 0 and establishes the ontological preconditions for field-level organization at Level 1.

The central claim is that time, space, energy, mass, and particle-like form should not be treated here as primitive givens. Within the

present framework, they arise as consequences of admissible structural differentiation, compensation, closure, and variational selection. In this sense, the paper proposes a topology of arising in which minimal action is reinterpreted as a structural selection principle operating within admissibility space. The result is a formal Level-0 framework for describing how the first stable units of organization become possible before standard physical ontology is introduced.

**Keywords:** admissibility structures, topology of arising, distinguishability, minimal action, structural closure, CRT-node, Level 0 ontology

## Introduction

Modern physics has achieved extraordinary descriptive and predictive power by formalizing already admissible regimes of reality: fields, particles, interactions, symmetries, and state spaces. Yet this success leaves a deeper ontological problem largely unresolved. Under what minimal structural conditions do such regimes become possible in the first place? This paper addresses that question by shifting the point of departure from already constituted physical entities to the prior admissibility conditions under which directional difference, compensation, stability, and localized form can arise.

The analysis therefore does not begin from spacetime, matter, or law. It begins from distinguishability within a proto-field of admissibility. The first task is to determine how a difference can persist under repetition without presupposing coordinates, metrics, trajectories, or physical motion. Once distinguishability becomes structurally preservable, compensation becomes necessary, and energy appears as the measure of that compensatory requirement. Once compensation becomes coherently repeatable, oscillatory stability becomes admissible. Once oscillatory modes enter closed resonant relation, the first nontrivial localized unit becomes possible. In this sequence, time, space, energy, mass, and localized form are reconstructed as arising outcomes of structural admissibility rather than as primitive inputs.

The mathematical apparatus introduced in this paper should be read in that sense. The operators, action principles, variational constraints, and closure relations used below are not asserted as established microscopic laws of nature at Level 0. They function as formal admissibility devices: compact structural expressions used to represent what must hold for specific configurations to become stable, repeatable, and ontologically selectable. Minimal

action is treated here as a structural selection principle that separates unstable deviation from admissible organization.

Within this framework, Level 0 is organized as a progression from distinguishability to closure. Phase A establishes the first contour of difference, directional operators, and compensatory measure. Phase B introduces stable oscillatory form and variational admissibility. The later segment of the same phase develops coherent triadic resonance as the first irreducible closure condition and yields the CRT-node as the first persistent localized unit. The final segment extends this logic to ensembles, where interacting nodes become the minimal collective quanta of structural organization and prepare the transition to Level 1. The purpose of the paper is therefore to reconstruct the ontological preconditions under which physical description itself becomes admissible.

## **Level 0, Phase A: The first ontological contour of distinguishability**

The initial problem at Level 0 is not interaction, motion, or substance, but distinguishability. Without a difference that can be preserved under repetition, no admissible comparison can stabilize, and without such comparison there can be no coordinates, measures, trajectories, or ordered change. Phase A therefore concerns the first ontological contour within which difference becomes structurally preservable. The issue at this stage is not the behavior of pre-existing entities, but the minimal condition under which any entity-like persistence could later become admissible.

Distinguishability is defined here as the capacity of a configuration to sustain a local difference relative to an otherwise undifferentiated background in a form that remains identifiable across admissible repetitions. This definition does not presuppose space, time, or metric structure. It presupposes only that comparison itself can be admitted in a way that preserves the identity of difference. Distinguishability is therefore not yet a relation between objects. It is the first structural asymmetry through which ontological articulation becomes possible.

## **The proto-field of admissibility**

Let  $\Omega$  denote the proto-field of admissibility.  $\Omega$  is neither a physical vacuum nor a material substrate. It is defined only by the absence of privileged

distinctions and by invariance under admissible transformations. Within  $\Omega$ , local deviations are possible, but position, direction, and magnitude are not yet given *a priori*.

A local deviation is denoted by  $\delta\Omega$ . Such a deviation does not yet constitute motion or energy. It introduces structural tension: the requirement that difference be accounted for within a field that otherwise contains no intrinsic preference.

## Compensation and the emergence of energy

Any admissible deviation  $\delta\Omega$  induces a requirement of compensation. Compensation is introduced here as a minimal structural condition for preserving admissibility under repeated comparison. Let  $\mathcal{C}$  denote the minimal compensation function required to restore structural coherence.

Energy is then defined as the structural measure of compensatory requirement:

$$E = \mathcal{C}(\delta\Omega). \quad (1)$$

At this stage, energy is neither a substance nor an externally supplied quantity. It is the measure of how strongly a configuration must compensate in order to preserve admissible distinguishability.

## The first operators of difference

Let  $F$  denote a localized admissible deviation within  $\Omega$ . The first admissible operations are operators of difference rather than analytical derivatives in the usual physical sense. Their role is to encode directional distinguishability under repetition.

Two primitive operators are introduced:  $\partial_t$ , representing sequential distinguishability and thereby opening the condition for temporal directionality, and  $\partial_x$ , representing directional distinguishability and thereby opening the condition for spatial localization. At this stage, these operators remain structural. They acquire analytical interpretation only later, once a more explicit geometrical organization becomes admissible.

## Aggregation of distortion

Let  $f$  denote the density of admissible distortion associated with  $F$ . The aggregation of distortion may then be represented symbolically as

$$E \approx \int f^2. \quad (2)$$

Here the integral sign denotes aggregation over distinguishable support rather than integration over an already constituted physical space. Spatial interpretation becomes admissible only after localization achieves structural stability.

## Transition from Phase A to Phase B: Closure of distinguishability

Once distinguishability, compensation, and directional operators enter a coherent and repeatable configuration, the system passes from Phase A to Phase B. This transition marks the first ontological closure: differences no longer appear as transient deviations alone, but as self-consistent structural relations capable of persistence under repetition.

## Stable oscillatory configuration

In Phase B, admissible deviations may organize into coherent oscillatory configurations. Let

$$\psi(x, t) = Ae^{i(kx - \omega t)} \quad (3)$$

denote a minimal stable compensatory form. At this stage, the expression does not yet represent a physical wave in the standard sense. It denotes the simplest oscillatory configuration capable of preserving admissible compensation under repeated distinguishability.

## Minimal action as an admissibility filter

Stable configurations are selected through a Lagrangian admissibility filter:

$$\mathcal{L} = \frac{1}{2} \eta^{\mu\nu} \partial_\mu \psi^* \partial_\nu \psi - \frac{1}{2} m^2 \psi^* \psi. \quad (4)$$

Here the pseudo-Euclidean signature is not yet assumed as a physical space-time background, but is introduced as the candidate minimal geometrical

constraint under which complex oscillatory compensation may remain variationally stable. Its necessity is now made explicit in the following proposition. The Lagrangian therefore functions at this stage as a structural selection device: it compactly expresses the conditions under which oscillatory configurations can persist as admissible rather than collapse as unstable deviations.

## Uniqueness of the minimal admissibility geometry

The introduction of a pseudo-Euclidean signature should not remain a purely formal choice. At the present level, it can be justified as the unique minimal geometrical class compatible with the admissibility requirements already established by distinguishability, compensation, and bounded oscillatory persistence. The issue is not which geometry later empirical physics will ultimately realize in full detail, but which signature is already forced by the minimal Level-0 conditions under which sequential and directional difference can jointly support stable oscillatory closure.

Let  $V$  be the minimal two-dimensional space generated by the two primitive admissibility directions already introduced, namely sequential distinguishability and directional distinguishability. Let  $g$  be a real symmetric bilinear form on  $V$ , and let the admissibility action be written in the general quadratic form

$$S[\psi] = \int dx dt \left( \frac{1}{2} g^{ab} \partial_a \psi^* \partial_b \psi - \frac{1}{2} m^2 \psi^* \psi \right), \quad (5)$$

where  $a, b \in \{0, 1\}$  correspond to the two primitive directions of admissible differentiation.

The admissibility requirements imposed on  $g$  are minimal.

1. *Nondegeneracy.* The two primitive directions must remain structurally independent. Therefore  $g$  must be nondegenerate.
2. *Minimal quadraticity.* The first admissible variational filter is assumed to be quadratic in first derivatives, since no higher-order geometrical structure has yet become admissible.
3. *Ordered asymmetry of directions.* Sequential distinguishability and directional distinguishability must not collapse into two geometrically equivalent axes, because the former opens ordered variation whereas the latter opens localization.

4. *Compensatory boundedness.* Admissible oscillatory modes must allow compensatory balance rather than unidirectional accumulation of all quadratic contributions.
5. *Localization compatibility.* The resulting oscillatory regime must admit bounded coherent configurations, so that later triadic closure and localized node formation remain structurally possible.

These conditions are sufficient to determine the signature class of  $g$ .

**Proposition.** Under the five admissibility conditions listed above, the bilinear form  $g$  must have indefinite signature. In two dimensions this implies that  $g$  has signature  $(1, 1)$ . Hence, by Sylvester's law of inertia, every admissible  $g$  is equivalent up to linear change of basis and overall scale to

$$g \sim \lambda \operatorname{diag}(-1, +1), \quad \lambda > 0. \quad (6)$$

**Proof.** Since  $g$  is a real symmetric nondegenerate bilinear form on a two-dimensional vector space, only three signature classes are possible:  $(2, 0)$ ,  $(0, 2)$ , or  $(1, 1)$ .

The definite cases  $(2, 0)$  and  $(0, 2)$  make the two primitive admissibility directions geometrically equivalent up to sign. In that situation the quadratic derivative term assigns the same sign to both axes, so the resulting variational structure does not distinguish between sequential ordering and directional localization at the level of signature. The action then admits only same-sign accumulation of derivative contributions. This is incompatible with the required compensatory balance between ordered variation and directional localization.

More explicitly, in the definite case the corresponding Euler–Lagrange operator has elliptic type, schematically of the form

$$\partial_0^2 \psi + \partial_1^2 \psi + m^2 \psi = 0 \quad (7)$$

up to an overall sign. Such a form does not provide the minimal ordered asymmetry required for one axis to function as the carrier of sequential distinction and the other as the carrier of localization. It therefore fails the admissibility condition that ordered differentiation and directional differentiation remain structurally distinct in role.

By contrast, the indefinite case assigns opposite sign to the two primitive directions. The Euler–Lagrange operator then takes hyperbolic form,

$$-\partial_t^2\psi + \partial_x^2\psi + m^2\psi = 0, \quad (8)$$

which preserves a minimal structural asymmetry between sequential and directional differentiation while still allowing them to enter a single coherent variational filter. The corresponding dispersion relation

$$m^2 = \omega^2 - k^2 \quad (9)$$

expresses precisely the compensatory difference required for admissible bounded oscillatory organization. This makes possible both variational stability and later localized closure.

Therefore the definite signatures are inadmissible, and the only remaining nondegenerate signature class is  $(1, 1)$ . This is exactly the pseudo-Euclidean class.  $\square$

The result should be read in a strictly structural sense. It does not yet claim the full empirical geometry of physical spacetime. It establishes only that, at Level 0, once minimal admissibility is required for ordered difference, compensatory oscillation, and localization-compatible boundedness, the unique minimal geometrical class is pseudo-Euclidean.

**Remark.** The proposition does not claim that the full empirical geometry of later physical theory has already been derived in complete form. Its scope is narrower and more precise. It establishes that, within the Level-0 admissibility framework, once one requires nondegenerate ordered distinction, compensatory variational stability, and localization-compatible boundedness, the minimal admissible metric class is uniquely hyperbolic. In this sense, pseudo-Euclidean structure appears here not as an imported spacetime assumption but as a derived structural condition of admissible oscillatory organization.

## Uniqueness of the minimal admissibility operator

Once the admissible metric class has been fixed, the next question is whether the corresponding differential operator also remains underdetermined. At Level 0, the answer is again negative. Under the same minimal admissibility assumptions, the second-order operator compatible with ordered distinction,



compensatory balance, and bounded oscillatory persistence is unique up to scale and linear coordinate choice.

Let the admissible quadratic action be written in the already established hyperbolic class:

$$S[\psi] = \int dx dt \left( \frac{1}{2} g^{ab} \partial_a \psi^* \partial_b \psi - \frac{1}{2} m^2 \psi^* \psi \right), \quad (10)$$

with  $g$  nondegenerate and of signature  $(1, 1)$ . Since  $g$  is symmetric, the associated second-order differential operator is

$$\mathcal{D}_g \psi = g^{ab} \partial_a \partial_b \psi. \quad (11)$$

At the minimal level of admissibility, no further structure is available from which additional derivative terms could legitimately arise. In particular, the following are excluded:

1. *First-order drift terms*  $b^a \partial_a \psi$ , because they would introduce a preferred propagation bias not grounded in the primitive distinction structure itself.
2. *Mixed non-symmetric terms*, because the admissibility functional is generated by a symmetric quadratic form.
3. *Higher-order derivatives*, because no higher-order geometric or configurational structure has yet become admissible at Level 0.
4. *Nonlinear corrections*, because nonlinear closure belongs only to the later CRT-specific regime and not to the first minimal oscillatory filter.

It follows that the minimal admissible operator must coincide with the second-order operator canonically induced by the admissible metric.

**Proposition.** Let  $g$  be a nondegenerate symmetric bilinear form of signature  $(1, 1)$  on the minimal admissibility space  $V$ . Then every minimal second-order linear admissibility operator compatible with the variational structure is equivalent, up to an overall nonzero scale and linear change of coordinates, to

$$\square = -\partial_t^2 + \partial_x^2. \quad (12)$$

**Proof.** Since  $g$  has signature  $(1, 1)$ , Sylvester's law of inertia guarantees the existence of a linear basis  $(t, x)$  in which

$$g^{ab} = \lambda \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \lambda \neq 0. \quad (13)$$

Therefore

$$\mathcal{D}_g = g^{ab} \partial_a \partial_b = \lambda (-\partial_t^2 + \partial_x^2). \quad (14)$$

The overall factor  $\lambda$  does not define a distinct operator class, since it can be absorbed into normalization of the action or field rescaling. Thus the operator is unique up to scale.

No other second-order linear operator belongs to the same minimal admissibility class. Any added first-order term would introduce an ungrounded directional drift; any higher-order term would presuppose additional admissible structure not yet established; and any nonlinear correction would belong to a later closure regime rather than to the minimal oscillatory level itself. Hence the unique minimal operator is the hyperbolic d'Alembertian class.  $\square$

This result is structurally important. It shows that once the pseudo-Euclidean admissibility class is derived, the corresponding minimal wave operator is not an additional postulate. It is the unique second-order variational operator compatible with that class. The Level-0 transition from primitive distinction to oscillatory dynamics therefore does not require importing the d'Alembertian from later physics. It arises as the minimal admissible operator already forced by the internal logic of the construction.

## Necessity of the minimal mass term

Once the admissible hyperbolic metric class and the corresponding minimal second-order operator have been fixed, the remaining question is whether the variational functional may consist of the derivative term alone, or whether an additional scalar closure term is already required at Level 0. At this stage, the issue is not empirical particle mass in the later physical sense. The issue is whether a purely kinetic admissibility functional is sufficient to select bounded oscillatory organization, or whether minimal structural closure already demands an internal scalar term.

Consider first the purely kinetic functional

$$S_0[\psi] = \int dx dt \frac{1}{2} g^{ab} \partial_a \psi^* \partial_b \psi. \quad (15)$$

Its Euler–Lagrange equation is

$$\square\psi = 0. \quad (16)$$

This equation admits propagating modes, but it does not by itself introduce any intrinsic internal scale separating merely propagating admissible difference from bounded oscillatory closure. The resulting dynamics remains scale-free with respect to internal localization: no configurational parameter yet distinguishes a mode that simply propagates from one that can serve as a candidate for later bounded resonant persistence.

At Level 0, however, admissibility requires more than mere propagation. The oscillatory regime must remain compatible with boundedness, compensatory closure, and later localized resonance. This requires a scalar term that functions as an internal closure parameter within the variational filter.

**Proposition.** Under the minimal Level-0 admissibility conditions, the quadratic variational functional cannot remain purely kinetic. The unique minimal local scalar correction compatible with the established structure is a term proportional to

$$\psi^*\psi. \quad (17)$$

Therefore the minimal admissibility functional must take the form

$$S[\psi] = \int dx dt \left( \frac{1}{2} g^{ab} \partial_a \psi^* \partial_b \psi - \frac{1}{2} m^2 \psi^* \psi \right), \quad (18)$$

with  $m^2$  acting as the minimal scalar closure parameter.

**Proof.** The correction term must satisfy five conditions.

1. *Locality.* It must depend only on the local field configuration already admissible at Level 0.
2. *Quadratic minimality.* It must remain within the same minimal quadratic variational class as the kinetic term.
3. *Scalar invariance.* It must not introduce any new preferred direction beyond those already encoded in the metric structure.
4. *Reality of the action.* It must preserve the real-valued character of the admissibility functional.

5. *Closure functionality.* It must provide an intrinsic internal scale capable of distinguishing bounded oscillatory organization from scale-free propagation.

Terms linear in  $\psi$  or  $\psi^*$  are excluded, because they depend on an arbitrary choice of phase or sign and therefore do not define a stable scalar closure contribution. First-order derivative corrections are excluded because they introduce directional drift rather than internal closure. Mixed derivative-free quadratic terms other than  $\psi^*\psi$  are excluded because, for a complex admissible field,  $\psi^*\psi$  is the unique quadratic local scalar that is basis-independent and real.

Therefore the only admissible minimal scalar correction is

$$-\frac{1}{2}m^2\psi^*\psi, \tag{19}$$

where the coefficient  $m^2$  enters as a structural parameter measuring the degree to which an oscillatory mode possesses internal closure against unrestricted dispersion.

The parameter  $m^2$  should not yet be read as empirical particle mass. At this stage it functions as the minimal internal closure invariant of the oscillatory regime. Only later, once localized resonant configurations and persistent node-like units become admissible, can this closure parameter be reinterpreted as the precursor of what physical theory will later describe as mass.  $\square$

This point is structurally decisive. The term  $m^2\psi^*\psi$  is not inserted here by analogy with familiar relativistic field theory. It is the unique minimal scalar closure term required once admissible dynamics must distinguish free propagation from bounded oscillatory organization. In this sense, the mass term is already demanded by the internal logic of admissibility before any later physical identification of particles or spacetime ontology.

## Discrete compensation scale

A scale parameter  $\hbar$  appears as a marker of discrete compensatory action, expressed through

$$E = \hbar\omega \tag{20}$$

and

$$p = \hbar k. \tag{21}$$

At this level,  $\hbar$  is neither assigned a numerical value nor treated as an already established fundamental constant. Its role is to mark the discreteness of admissible compensation in oscillatory regimes, that is, the fact that stable compensatory organization does not vary continuously without constraint, but becomes selectable only in structurally bounded increments.

## Emergent mass as an integrability condition

The corresponding action is defined as

$$S[\psi] = \int dx dt \left( \frac{1}{2} \eta^{\mu\nu} \partial_\mu \psi^* \partial_\nu \psi - \frac{1}{2} m^2 \psi^* \psi \right), \quad (22)$$

and admissible stationarity is imposed by

$$\delta S = 0. \quad (23)$$

At this stage, the action is not introduced as a ready-made physical law. Its derivative part is fixed by the unique minimal admissibility operator, while its scalar term is fixed by the requirement that oscillatory admissibility not remain scale-free but admit an internal closure parameter. The resulting action therefore functions as the minimal complete variational formalism under which oscillatory configurations can remain structurally stable under variation.

Under this condition, variation with respect to  $\psi^*$  yields a Klein–Gordon-type form

$$\square \psi + m^2 \psi = 0, \quad (24)$$

where the operator  $\square$  is now understood not as an imported spacetime primitive, but as the unique minimal second-order admissibility operator selected by the hyperbolic metric class:

$$\square = -\partial_t^2 + \partial_x^2. \quad (25)$$

The operator  $\square$  should therefore be read here as the unique minimal variational expression of structured temporal and directional difference within the admissible hyperbolic class derived above, rather than as the already constituted spacetime wave operator of later physical theory.

The scalar closure parameter introduced above then appears, at the level of admissible mode solutions, as an integrability condition for oscillatory organization, expressed in normalized units as

$$m^2 = \omega^2 - k^2. \quad (26)$$

In this form,  $m^2$  measures not yet a substance-like physical mass, but the degree to which the oscillatory configuration possesses internally closed admissible structure rather than unrestricted propagative dispersion.

## Lagrangian of admissibility

The admissibility Lagrangian introduced above defines the minimal structural filter through which oscillatory configurations become selectable as stable under admissible transformations. Its role at this stage is not to state a physical law, but to compactly express the variational condition under which compensatory organization can persist without structural collapse.

The condition

$$\delta S = 0 \tag{27}$$

therefore functions as a criterion of structural stationarity. It selects configurations that remain coherent under admissible variation and excludes those that fail to preserve oscillatory closure.

## First ontological closure at Level 0

At this point, the minimal system of distinguishability, compensation, directional operators, oscillatory form, and variational selection reaches its first ontological closure. The triad  $(\partial_t, \partial_x, E)$ , together with its organization through  $\psi$  and  $\mathcal{L}$ , defines the first self-consistent contour in which difference is no longer merely induced, but can be preserved, filtered, and stabilized as a repeatable structural configuration.

This marks the completion of the first closure segment of Phase B at Level 0 and establishes the entry conditions for triadic resonance within the same phase.

- The first directional differentiation through which temporal and spatial order become admissible.
- The first compensatory metric through which deviation is structurally measurable as energy.
- The first stable oscillatory configuration capable of preserving admissible compensation under repetition.

- The first variational constraint through which admissible configurations are structurally filtered.
- The first selection principle through which stable closure becomes distinguishable from unstable deviation.

## Level 0, Phase B: Three-wave resonance as a condition of stable configuration

Phase B of Level 0 concerns the emergence of the first structurally closed and self-supporting multi-mode configurations. Its first segment establishes minimal oscillatory closure through  $(\partial_t, \partial_x, E)$ ,  $\psi$ , and  $\mathcal{L}$ . The second segment introduces triadic resonance as a higher-order condition of internal coherence, under which multiple oscillatory modes can enter a self-supporting relation and thereby sustain structural stability without external stabilization.

Assume that a stable oscillatory configuration

$$\psi(x, t) = Ae^{i(kx - \omega t)} \quad (28)$$

has become admissible under the pseudo-Euclidean constraint. This configuration minimizes structural action

$$S[\psi] = \int dx dt \left( \frac{1}{2} \eta^{\mu\nu} \partial_\mu \psi^* \partial_\nu \psi - \frac{1}{2} m^2 \psi^* \psi \right) \quad (29)$$

and thereby defines the first organized oscillatory axis within the field  $\Omega$ . Such a mode remains stable only within a bounded range of parameters  $(k, \omega, \hbar)$ , indicating that stability is already a selective structural condition rather than a generic consequence of oscillation alone.

## Coherent resonant closure

### Why dyadic closure is insufficient

The transition from admissible oscillation to localized structural closure requires more than coexistence of multiple modes. It requires the first configuration in which internal persistence is not reducible to a single pairwise relation or to simple mirror symmetry. This point is decisive, because the present framework does not seek the first interaction, but the first irreducible closed contour capable of supporting bounded self-maintaining organization.

To make this precise, consider first the dyadic case. Let two admissible oscillatory modes be given:

$$\psi_1(x, t) = A_1 e^{i(k_1 x - \omega_1 t)}, \quad \psi_2(x, t) = A_2 e^{i(k_2 x - \omega_2 t)}. \quad (30)$$

If one imposes compensation conditions

$$k_1 + k_2 = 0, \quad \omega_1 + \omega_2 = 0, \quad (31)$$

then the resulting composite configuration is

$$\Psi_2(x, t) = \psi_1 + \psi_2. \quad (32)$$

This configuration can exhibit balance, interference, and bounded oscillatory superposition, but it does not yet constitute the first irreducible closure. The reason is structural. A dyadic relation remains exhaustively describable by one mutual coupling axis: each mode is paired only with the other, and the entire configuration is reducible to a single compensatory opposition. No third relation is present that could stabilize the pair through an independent closure path. The dyad therefore remains structurally decomposable into a binary symmetry rather than a genuinely closed contour.

In graph-theoretic terms, the dyadic case yields only one edge class of mutual dependence. It may produce opposition, reflection, or standing balance, but not the first minimal cycle with nontrivial relational overdetermination. In topological terms, the dyad defines a segment, not a loop. In variational terms, it admits balanced propagation, but not the first internally overconstrained contour whose stability depends on more than one independent relation.

**Proposition.** A dyadic admissible oscillatory configuration is insufficient to produce the first irreducible localized closure. The minimal admissible contour of irreducible closure requires three coherently related modes.

**Proof.** A minimal irreducible closure must satisfy three conditions.

1. *Closed relational support.* Each constituent must be supported not only by one compensatory partner, but by a closure structure that remains nontrivial under decomposition.



2. *Relational overdetermination.* The full configuration must contain more constraints than can be reduced to one binary opposition.
3. *Minimal loop structure.* The configuration must realize the first genuine cycle rather than a reversible pair.

A dyad fails all three conditions. Its support is binary only; its constraint structure is exhausted by one pairwise relation; and its topology is that of a reversible segment rather than a loop. If one constituent is removed, no residual closure remains. Even when the dyad is balanced, its stability is still reducible to one opposition relation and therefore does not yet define the first irreducible contour.

Now consider three modes  $\psi_1, \psi_2, \psi_3$  satisfying

$$k_1 + k_2 + k_3 = 0, \quad \omega_1 + \omega_2 + \omega_3 = 0. \quad (33)$$

The resulting configuration is no longer exhausted by one binary opposition. Each mode participates in a closure relation whose persistence depends on the other two. The system therefore contains the first minimal loop of compensatory support. Removal of any one mode destroys the closure class itself rather than merely weakening one pairwise interaction.

Hence the triadic case is the first admissible configuration in which bounded oscillatory organization becomes irreducibly closed. The dyadic case may realize balance, but only the triadic case realizes the first nontrivial contour of closure.  $\square$

This is the structural reason why the present framework passes from admissible wave coherence to triadic resonant closure rather than stopping at paired oscillatory compensation. Two modes are sufficient for opposition. Three modes are required for the first irreducible contour.

## Configurational density as the first scalar invariant of closure

Once triadic resonant closure has been identified as the first irreducible admissible contour, the next structural question is how such a contour can be represented by a scalar quantity that remains stable under admissible transformations of description. At this stage, one does not yet seek a full object ontology or a fully developed localization theory. One seeks the first invariant

measure that distinguishes mere coexistence of oscillatory components from a genuinely concentrated closed configuration.

The requirement is minimal. The quantity sought must satisfy four conditions.

1. *Scalarity.* It must be invariant under admissible linear re-description of the composite field and must not depend on choosing one constituent mode as privileged.
2. *Positivity.* It must distinguish the presence of concentrated coherent support from trivial cancellation or signed compensation.
3. *Local aggregability.* It must be definable pointwise and then aggregable over the support of the configuration.
4. *Closure sensitivity.* Its value must increase when resonant coherence becomes concentrated and must collapse when closure is destroyed.

Let

$$\Psi(x, t) = \psi_1(x, t) + \psi_2(x, t) + \psi_3(x, t) \quad (34)$$

be the composite resonant configuration. At each admissible point of support, the simplest scalar satisfying the four conditions above is

$$\rho_\Psi(x, t) = \Psi^*(x, t)\Psi(x, t) = |\Psi(x, t)|^2. \quad (35)$$

This quantity is scalar with respect to phase-preserving linear field description, nonnegative by construction, locally defined, and directly sensitive to concentration of coherent support. It is therefore the first admissible local invariant of resonant closure.

**Proposition.** For a minimal complex oscillatory closure regime, the quantity  $|\Psi|^2$  is the unique minimal local scalar density that is quadratic, non-negative, and invariant under admissible phase-preserving re-description of the composite configuration.

**Proof.** At the present level, the invariant must be built only from the admissible field  $\Psi$  and its complex conjugate  $\Psi^*$ , since no higher-order geometrical or internal object structure has yet been established.

A linear expression in  $\Psi$  or  $\Psi^*$  is excluded because it is not phase-invariant and can change sign under admissible re-description. A derivative-dependent scalar is excluded because the goal at this point is not yet to measure propagation or curvature, but concentration of already formed closure. Any higher-order polynomial is nonminimal. Therefore the first admissible candidate must be quadratic and derivative-free.

For a complex field, the only quadratic local scalar with these properties is

$$\Psi^*\Psi. \quad (36)$$

It is real, nonnegative, local, and independent of arbitrary phase choice. Hence it is the unique minimal scalar density of the closed configuration.  $\square$

The corresponding aggregated quantity is then

$$\Gamma(t) = \int \Psi^*(x, t) \Psi(x, t) dx. \quad (37)$$

This integral should still be read in the Level-0 sense: not yet as integration over an already constituted physical space, but as aggregation over distinguishable support within which closure has become localized strongly enough to be coherently counted.

In this form,  $\Gamma(t)$  is the first scalar invariant of localized closure. It measures neither matter density nor probability in the later physical sense, but the degree to which the closed resonant contour has become concentrated into bounded coherent support. Configurational density is therefore the first scalar bridge from triadic closure to localized node formation.

Let three independent oscillatory deviations therefore become admissible within the ontological field as the first minimal configuration capable of irreducible closure, expressed as

$$\psi_1(x, t) = A_1 e^{i(k_1 x - \omega_1 t)}, \quad \psi_2(x, t) = A_2 e^{i(k_2 x - \omega_2 t)}, \quad \psi_3(x, t) = A_3 e^{i(k_3 x - \omega_3 t)}. \quad (38)$$

These deviations do not yet constitute particles or fully formed fields. They are distinct oscillatory modes admitted by the underlying configuration of structural compensation.

A coherent resonant closure becomes possible only if the following conditions are satisfied:

$$k_1 + k_2 + k_3 = 0 \quad (39)$$

and

$$\omega_1 + \omega_2 + \omega_3 = 0. \quad (40)$$

These relations define the minimal conditions of phase closure and compensatory balance. They eliminate net drift and thereby ensure that the composite configuration can remain bounded, localized, and structurally coherent under repetition.

The closure conditions imply three structural properties. First, phase closure preserves a bounded internal phase relation across the composite configuration. Second, compensatory cancellation suppresses global propagation and thereby prevents unbounded dispersion. Third, mutual stabilization becomes possible because each oscillatory mode contributes to the persistence of the other two through the nonlinear response of the field.

### **The coherent resonant triad**

The minimal configuration satisfying these closure conditions is termed a Coherent Resonant Triad. Its minimality is structural rather than merely numerical: it is the first configuration in which irreducible closure is realized. The triad is neither an object nor a mere sum of waves. It is a configurational closure in which three admissible oscillatory modes enter a self-supporting resonance loop. At this point, complexity appears as the irreducibility of internal coupling.

Even at this minimal level, the triadic configuration introduces the first nontrivial irreducibility of organized internal dynamics, functionally analogous to the increase in complexity associated with three-body organization. It therefore places a nontrivial configurational load on the admissibility structure and marks the first transition from linear superposition to genuinely complex closure.

### **Structural closure and emergent node**

Within the present classification of systemic complexity, a coherently closed triad corresponds to the minimal case of three-body complexity  $\mathcal{C}_3$ . The internal dynamics of such a configuration cannot be fully reduced through dimensional decoupling, and its evolution retains intrinsic ambiguity constrained only by admissibility conditions.

The composite configuration is expressed as

$$\Psi(x, t) = \psi_1(x, t) + \psi_2(x, t) + \psi_3(x, t). \quad (41)$$

This expression should not be read as a merely linear sum. It denotes a resonantly locked composite structure whose admissibility is maintained through closure of phase, compensatory balance, and bounded support.

The corresponding configurational density is then obtained as the first aggregated scalar invariant of the closed resonant configuration:

$$\Gamma(t) = \int \Psi^*(x, t) \Psi(x, t) dx. \quad (42)$$

## Why the CRT-node has object status

At this point, the question is no longer whether a closed resonant contour exists, but whether such a contour has crossed the ontological threshold required for objecthood. Within the present framework, object status is not assigned merely because a configuration is complex, dense, or dynamically interesting. It is assigned only when a configuration becomes sufficiently bounded, invariant-bearing, and persistence-capable to function as a distinguishable carrier under admissible perturbation.

This criterion is consistent with the general SOMA classification of type roles, where an object is defined as a persistent configuration stabilized by internal coherence and admissible constraints, remaining identifiable under perturbations within a defined tolerance. The issue here is therefore whether the resonant triadic closure already satisfies the minimal Level-0 version of that condition.

The answer depends on three structural properties.

1. *Bounded support.* The configuration must occupy a distinguishable and limited support region rather than dissolve into unrestricted propagation.
2. *Invariant concentration.* The configuration must possess a scalar invariant of coherence, here given by  $\Gamma$ , that remains meaningful across admissible re-description.
3. *Perturbative persistence.* The configuration must remain identifiable under small admissible perturbations, rather than collapsing immediately when exact resonance is slightly disturbed.

If these three conditions are met simultaneously, the configuration is no longer merely a pattern distributed across a carrier. It becomes the first admissible carrier in its own right.

**Proposition.** A closed resonant configuration satisfying bounded support, nontrivial configurational density, and perturbative persistence qualifies as the first object-like carrier at Level 0.

**Proof.** A pattern is defined by reproducible relational organization, but it does not yet require bounded identity as a distinguishable unit. An object, by contrast, requires persistence as a coherent carrier. Therefore the transition from pattern to object occurs when relational closure is no longer merely repeatable, but becomes locally stabilized strongly enough to be reidentified across admissible variation.

For the closed resonant configuration  $\Psi$ , bounded support ensures that the configuration is localized rather than merely extended. The scalar invariant

$$\Gamma = \int \Psi^*(x, t) \Psi(x, t) dx \quad (43)$$

ensures that coherence is not described only qualitatively, but possesses a stable aggregated measure. Perturbative persistence ensures that the configuration is not an exact fine-tuned singularity, but a structurally stable unit under small admissible deformation.

If any one of these three conditions fails, object status also fails. Without bounded support, the configuration remains an extended pattern. Without invariant concentration, it lacks a stable coherence measure. Without perturbative persistence, it remains a transient coincidence rather than an admissible carrier. Only their conjunction is sufficient.

Therefore, once triadic closure yields a bounded coherent support whose concentration is measurable by  $\Gamma$  and whose identity survives small admissible perturbation, the resulting configuration crosses the minimal threshold from pattern to object.  $\square$

The CRT-node should therefore be read as the first object-like arising of Level 0. It is not yet a classical particle, not yet an empirically established microscopic constituent, and not yet a later-level physical body. But it is already more than a pattern. It is the first bounded coherent carrier whose identity is maintained by internal closure strongly enough to justify object status within the ontology of admissibility.

Here the integral denotes aggregation over localized distinguishable support rather than integration over a fully constituted physical space. What is being aggregated is not yet matter or probability, but the minimal scalar density of resonant closure derived above. It is precisely this aggregated invariant that allows the closed contour to be treated not only as a pattern of relations but as a candidate carrier of bounded coherent identity. Spatial interpretation becomes admissible only after localization has stabilized. Together with the pseudo-Euclidean signature  $\eta$ , this density fixes the closed configuration as a unified node within the chosen admissibility structure.

## Frame-stable measure of the node

The configurational density

$$\Gamma = \int \Psi^*(x, t) \Psi(x, t) dx \quad (44)$$

functions as the first translation-stable scalar descriptor of node coherence within the chosen structural frame, because it is already the minimal invariant that measures concentration of closed resonant support and thereby underwrites the object-status of the emerging node.

## Structural properties of the node

A node formed through coherent triadic resonance exhibits the following structural properties, which jointly justify its status as the first object-like carrier at Level 0:

- It remains bounded under admissible evolution and resists dispersive loss of coherence.
- It preserves temporal stability through sustained phase relations.
- It remains structurally closed, with each component contributing to the persistence of the others.
- It exhibits internal compensatory symmetry, expressed by the condition  $k_1 + k_2 + k_3 = 0$ .
- It establishes the first localized foundation for further organizational layers capable of accumulating additional internal degrees of freedom.

## Ontological interpretation and relation to field theory

In quantum field theory, a bound state is understood as a stable configuration maintained through sustained interaction among multiple excitations. The coherent resonant triad plays an analogous ontological role within the present framework. Resonance functions as the mechanism of self-binding,  $\Omega$  provides the admissibility background within which such closure can be sustained, and  $\Gamma$  serves as a coarse scalar descriptor of the node's concentration, coherence, and persistence.

The point of this analogy is structural rather than identificatory. The triad is not introduced here as a standard bound state of an already constituted physical field theory. It is introduced as the minimal admissible configuration in which resonant closure, bounded support, and internal persistence coincide strongly enough for a unified node to become ontologically selectable.

## Second cycle of arising

This stage introduces a second cycle of arising. The first cycle establishes admissible difference through  $\partial_t$ ,  $\partial_x$ , and  $E$ , culminating in the stable oscillatory mode  $\psi$ . The second cycle integrates multiple oscillatory modes  $\psi_1$ ,  $\psi_2$ , and  $\psi_3$  into a unified resonant configuration  $\Psi$ , whose bounded coherence yields a node characterized by  $\Gamma$ .

Formally, this sequence may be expressed as

$$\Omega \rightarrow \psi \rightarrow \Psi \rightarrow \Gamma \rightarrow \text{node}. \quad (45)$$

This marks the transition from oscillatory admissibility to structurally closed form.

## Completion of Phase B at Level 0

The formation of a coherent resonant triad completes Phase B of Level 0. At this point, the system passes from isolated oscillatory admissibility to the minimal condition of structural embodiment. Further evolution no longer concerns fluctuations alone, but the interaction, persistence, and integration of bounded forms. This establishes the ontological foundation for particle-like organization, field-like interaction structures, and configurations with internal structure. At this stage, arising acquires localized form.



## Level 0, Phase C: Lagrangian of the resonant node and higher configurational units

Phase C of Level 0 concerns the formal stabilization and integration of the coherent resonant node. Whereas Phase B establishes a self-supporting triadic closure, Phase C introduces the minimal formal machinery required to describe the node's internal coherence, admissible interactions, and further integration into higher-order configurational structures.

### Formalization of the coherent resonant triad

Let  $\psi_i(x, t)$  denote three coherent oscillatory modes satisfying the closure conditions  $k_1 + k_2 + k_3 = 0$  and  $\omega_1 + \omega_2 + \omega_3 = 0$ . These conditions define phase closure and compensatory balance, thereby eliminating net drift and ensuring that the resulting configuration remains localized, bounded, and internally self-consistent.

The composite configuration of the node is expressed as

$$\Psi = \sum_{i=1}^3 A_i e^{i(k_i x - \omega_i t)}. \quad (46)$$

This expression should not be read as a mere superposition of independent waves. It denotes a resonantly locked composite configuration whose internal coherence cannot be reduced to the isolated contribution of its components.

### Admissibility Lagrangian of the resonant node

Within a 1+1 dimensional admissibility structure with pseudo-Euclidean signature  $\eta^{\mu\nu} = \text{diag}(-1, +1)$ , the generalized Lagrangian of the coherent resonant triad is written as

$$\mathcal{L}_{CRT} = \frac{1}{2} \eta^{\mu\nu} \partial_\mu \Psi^* \partial_\nu \Psi - \frac{1}{2} M^2 \Psi^* \Psi - \lambda(\psi_1 \psi_2 \psi_3 + \text{c.c.}). \quad (47)$$

This Lagrangian is introduced as a structural admissibility filter selecting bounded and self-supporting configurations of the resonant node.

Here  $M^2$  denotes the emergent mass of the node as a configurational parameter, while  $\lambda$  denotes the strength of nonlinear resonant coupling. The cubic term  $\psi_1 \psi_2 \psi_3$  encodes the minimal nonlinear coherence required for

structural self-binding, and the complex conjugate term enforces symmetry of interaction within the closure relation.

The cubic resonant term  $\lambda\psi_1\psi_2\psi_3$  introduces nonlinear coherence as a constitutive feature of the configuration, enabling the formation of an ontologically closed unit rather than a transient fluctuation. At this stage, arising produces not only localized form, but the first structural individuality of form.

With this, the logic of single-node closure is complete. The next question is whether such localized units can enter a genuinely collective admissibility regime.

## Equation of admissible motion

The admissibility action

$$S = \int dx dt \mathcal{L}_{CRT} \quad (48)$$

yields, under the stationarity condition  $\delta S = 0$ , an effective nonlinear equation of admissible motion of the form

$$\eta^{\mu\nu} \partial_\mu \partial_\nu \Psi + M^2 \Psi + \lambda R_{eff}^*(\psi_1, \psi_2, \psi_3) = 0, \quad (49)$$

where

$$R_{eff}(\psi_1, \psi_2, \psi_3) = \psi_1\psi_2 + \psi_2\psi_3 + \psi_3\psi_1. \quad (50)$$

This equation defines the effective admissible dynamics of the resonant node within the pseudo-Euclidean structure. It should not be read as a fundamental equation of microscopic physics, but as a compact formal expression of the conditions under which resonant closure can remain bounded and self-supporting under admissible variation.

The resulting dynamics admits bounded and stable solutions. Bounded concentration is not imposed externally, but arises from the balance between linear admissibility constraints and nonlinear resonant self-coupling.

## Emergent transition from node to particle-like unit

A coherently stabilized resonant node behaves, at the level of admissibility, as a particle-like unit in 1+1 dimensions. It represents a persistent bounded excitation, carries an emergent configurational parameter  $M$ , exhibits discrete

internal modes, and functions as a building block for subsequent organizational layers. This does not assert the existence of physical particles. It establishes the ontological precondition under which particle-like entities can later become admissible.

## **Ensemble of resonant nodes and meta-Lagrangian**

### **Why an ensemble of nodes defines a new collective regime**

The passage from a single resonant node to a plurality of interacting nodes should not be interpreted as a merely quantitative enlargement of the same ontological class. A collection becomes a new collective regime only when the admissibility of the whole is no longer reducible to the persistence of each node taken independently. At that point, stability, selection, and boundedness depend on inter-node relations as such, and not only on the internal coherence of the constituent nodes.

This distinction is decisive for the completion of Level 0. A single CRT-node is already an object-like carrier. But Level 0 is not complete until such carriers can enter a stable collective organization whose admissibility introduces a new layer of structural constraints. The issue is therefore not multiplicity alone, but the first emergence of irreducible collective admissibility.

Let  $\{\Psi_a\}_{a=1}^N$  denote a family of localized resonant nodes. If these nodes merely coexist without structurally relevant coupling, then the total configuration remains reducible to a disjoint collection of node-level objects. No new regime is obtained. By contrast, if the nodes enter bounded interaction through admissible coupling terms  $V_{ab}(\Psi_a, \Psi_b)$ , then the stability of the total configuration depends on both node-internal closure and inter-node compatibility.

**Proposition.** A family of CRT-nodes defines a new collective admissibility regime if and only if the stability of the total configuration cannot be reduced to the sum of independently admissible node persistences, but depends essentially on bounded inter-node coupling.

**Proof.** Suppose first that the nodes are uncoupled. Then the total admissibility functional decomposes as

$$S_{\text{tot}} = \sum_{a=1}^N S[\Psi_a]. \quad (51)$$

In that case, the admissibility of the whole is exhausted by the admissibility of each node separately. Removal or perturbation of one node changes the total sum, but does not change the structural admissibility class of the remaining configuration. Therefore no genuinely new collective regime exists.

Now suppose that bounded interaction terms are present, so that

$$S_{\text{tot}} = \sum_{a=1}^N S[\Psi_a] + \sum_{a<b} V_{ab}(\Psi_a, \Psi_b). \quad (52)$$

Then the admissibility of the total configuration depends not only on the persistence of each node, but on the compatibility structure among nodes. Removal or perturbation of one node may now alter the admissibility of the others by changing the interaction architecture itself. The collective configuration is therefore no longer reducible to a disjoint sum of local persistences.

This establishes a new regime of structural organization. The object of admissibility is no longer the single node alone, but the coupled ensemble as a bounded collective whole. Its coherence depends on inter-node relations that are not present at the isolated node level. Hence the ensemble is ontologically distinct from mere multiplicity.  $\square$

The consequence is straightforward. Once localized nodes enter bounded admissible coupling, the resulting whole becomes the first collective object-regime of Level 0. This is why the transition to an ensemble is not a numerical afterthought. It is the first case in which structural admissibility itself becomes collective.

## Completion of Level 0 and preparation for Level 1

Level 0 is not complete merely when the first bounded coherent carrier appears. It is complete only when such carriers enter a stable collective admissibility regime and thereby support a higher-level redescription. The relevant question is therefore no longer whether localized nodes exist, but whether their collective organization already functions as the minimal ontological bridge to a new carrier domain.

This condition is necessary because a new level of organization cannot begin from isolated persistence alone. It requires a regime in which already constituted carriers become admissible as elements of a new interaction domain, so that the next layer arises not directly from primitive oscillatory closure but from collective configurations of previously constituted units. In this sense, the ensemble is the first admissible completion of Level 0: it is the point at which localized units cease to be only the outcome of closure and become the basis of a new interaction domain.

**Proposition.** The ensemble of resonant nodes completes Level 0 if and only if it satisfies the following three conditions:

1. *Carrier closure.* The relevant units of organization are no longer primitive oscillatory modes but already constituted nodes  $\Psi_a$ .
2. *Collective admissibility.* Stability and selection depend on bounded inter-node coupling and not only on isolated node persistence.
3. *Effective re-describability.* The collective configuration admits a higher-level description in which the ensemble behaves as a new carrier domain for subsequent field-like organization.

**Proof.** If only the first condition is satisfied, then Level 0 has produced object-like units but has not yet generated a new collective regime. The result remains a set of isolated closures.

If the first and second conditions are satisfied, then the system has entered a collective admissibility regime. Node persistence is no longer sufficient by itself; bounded interaction architecture becomes part of the admissibility conditions. This already yields the first collective organization internal to Level 0.

However, Level 0 is not fully complete until the third condition is also satisfied. A new level cannot begin from a configuration that remains describable only as a finite list of interacting lower-level units. It begins only when those units admit a coherent effective redescription as elements of a new carrier domain. At that point, the collective whole is no longer treated merely as an ensemble in extension, but as the ontological basis of an emergent field-like regime.

Thus the ensemble completes Level 0 precisely when localized nodes become collectively admissible and simultaneously support an effective higher-level representation. This is the first point at which the ontology can pass from bounded node organization to field-level organization without returning to primitive distinguishability.  $\square$

The structural significance of this result is decisive. The ensemble is not simply the last and largest object of Level 0. It is the closure of the Level-0 cycle itself. Primitive distinction generated oscillatory admissibility; oscillatory admissibility generated irreducible resonant closure; resonant closure generated localized nodes; and node coupling generated the first collective carrier regime. Only then does the ontology possess a sufficiently stabilized basis for introducing an effective field of interactions among newly constituted units.

For this reason, the transition to Level 1 should not be read as a direct continuation of wave dynamics. It is a transition from collective node organization to an effective carrier domain. What Level 1 receives is not the primitive oscillatory field of Phase B, but the field-like redescription of ensemble-quanta already constituted through the completion of Level 0.

Let  $\{\Psi_a\}$  therefore denote a set of localized resonant nodes already functioning as the first collectively admissible carriers of completed Level-0 organization. An admissibility meta-Lagrangian for such an ensemble may then be introduced as

$$\begin{aligned} \mathcal{L}_{meta} = \sum_a \left[ \frac{1}{2} \eta^{\mu\nu} \partial_\mu \Psi^{a*} \partial_\nu \Psi^a - \frac{1}{2} M_a^2 \Psi^{a*} \Psi^a \right. \\ \left. - \lambda_a (\psi_1^a \psi_2^a \psi_3^a + \text{c.c.}) \right] + \sum_{a < b} V_{ab}(\Psi^a, \Psi^b). \end{aligned} \quad (53)$$

This expression does not yet constitute a fundamental many-body field theory. It serves as a structural admissibility formalism for describing how multiple resonant nodes can coexist, remain internally stabilized, and enter bounded interaction within the same pseudo-Euclidean organization.

The interaction terms  $V_{ab}$  are restricted to forms that preserve scalar character under the pseudo-Euclidean symmetry constraints adopted in the present formalism. Under this restriction, the ensemble can be treated not as a mere aggregate, but as an ontological graph of localized nodes whose collective admissibility depends on bounded interaction architecture. Each node retains its internal closure while entering external relations with other

nodes through admissible interaction channels, and it is precisely this coupling structure that defines the new collective regime.

As the number of nodes  $N$  increases, the number of possible interacting pairs and higher-order groupings grows combinatorially. This growth does not merely enlarge the configuration space. It marks the entry into a genuinely collective admissibility regime, because stability is now selected not only at the node level but at the level of the interaction architecture itself. The resulting structural constraints determine which multi-node patterns can remain admissible, bounded, and mutually coherent.

## Growth of configurational complexity

The transition from Phase B to Phase C corresponds not only to a shift in configurational complexity class, but to a shift from local closure to explicitly collective admissibility architecture. The system passes from single-mode oscillatory admissibility to triadic closure  $\mathcal{C}_3$ , and from there toward ensembles  $\mathcal{C}_n$  of interacting resonant nodes. With this shift, stability can no longer be characterized only in terms of local closure. It also requires selection principles that restrict which higher-order interaction patterns remain structurally admissible.

Resonant nodes are therefore no longer to be treated as isolated localized units alone. Within Phase C, they function as vertices in an ontological graph, endowed with metric relations, interaction links, and structural memory encoded in their configurational parameters. At this point, admissibility has become explicitly collective: what is selected is not only which nodes persist, but which coupled configurations of nodes can persist together.

## Ontological iteration of phases

At this level, the process of arising becomes explicitly iterative. Each phase introduces a new admissibility function and thereby redefines the structural conditions under which the next configurational layer can become admissible. The ontological action over configurations may be expressed as

$$S = \int dx dt \sum_n \mathcal{L}_{meta}^{(n)}(\Psi^{(n)}), \quad (54)$$

where  $n$  indexes layers of organization rather than temporal steps.

The cycle of arising is formalized as

$$\nabla \rightarrow C_1 \rightarrow C_2 \rightarrow C_3 \rightarrow \Delta \rightarrow \Omega^* \rightarrow \nabla'. \quad (55)$$

Here  $C_i$  denote successive configurational classes,  $\Delta$  denotes a critical admissibility threshold, and  $\Omega^*$  denotes the updated field of admissibility within which a new cycle of structural differentiation can begin. The process is therefore iterative not because the same state simply repeats, but because each completed layer transforms the admissibility conditions of the next.

## Completion of Phase C at Level 0

Phase C completes the transformation of admissible oscillatory configurations into structurally integrated units. The system passes from isolated resonant nodes to networks of interacting nodes governed by admissibility constraints. At this point, arising no longer concerns fluctuation alone, but the persistence, interaction, and integration of bounded configurational units. What becomes possible here is the collective carrier regime whose full ontological closure is completed in Phase D, where the ensemble is no longer only a network of nodes but the first fully constituted collective quantum of structural organization.

## Level 0, Phase D: Ensemble as the first collective quantum of structural organization

At this stage, four symbols are used with strict ontological separation.  $\Psi_a$  denotes a previously formed CRT-node treated as a bounded configurational unit.  $\phi_p(x, t)$  denotes a local internal mode associated with a node within an ensemble-level coupling structure. The term ensemble-quantum denotes the minimal collective unit generated by stable coupling among multiple nodes.  $\Phi(X, t)$  denotes the effective field introduced only at the next ontological level, where ensemble-quanta become the new basic carriers.

Phase D completes Level 0 by introducing the ensemble as the first fully constituted collective unit of structural organization. What was prepared in Phase C as a collective admissibility regime is here closed as an ontologically stabilized carrier class, capable of supporting the transition to effective field-level organization. At this stage, the resonant nodes formed in Phase C are no longer treated as isolated localized entities. Their coupling gives rise to



collective modes whose stability cannot be reduced to the properties of single nodes taken independently.

Each node carries a localized internal field  $\phi_p(x, t)$ . Coupling among nodes through a structural interaction parameter  $\kappa$  generates a collective oscillatory pattern. This pattern constitutes a single ensemble-quantum at the structural level. The term “ensemble-quantum” therefore denotes an indivisible configurational whole with its own stable dynamics, rather than a particle or an excitation of a microscopic physical field.

## Multiplicity of structural quanta under Lagrangian variation

Different admissible structural quanta arise depending on the form of the admissibility Lagrangian. For example, a Lagrangian of the form

$$\mathcal{L} \propto \frac{1}{2}(\partial\phi)^2 - \frac{1}{2}M^2\phi^2 - \frac{\lambda}{4}\phi^4 - \frac{g}{3}|\phi|^3 \quad (56)$$

admits a specific family of stable ensemble patterns. Modifying this structure through higher-order terms  $\phi^n$ , graph-dependent couplings, or nonlocal contributions yields qualitatively different classes of stable configurations.

Each admissible Lagrangian therefore defines its own family of ensemble-quanta. Structural diversity at this stage is not incidental. It follows directly from the way admissibility constraints are encoded in the variational structure. Different ensemble-quanta correspond to different modes of collective stability, boundedness, and internal organization.

## Alternative admissible forms in pseudo-Euclidean structure

The configurations discussed so far arise from the minimal triad  $\{\partial_t, \partial_x, E\}$ . However, even within a 1+1 dimensional pseudo-Euclidean structure, the admissibility field  $\Omega$  admits multiple locally stable oscillatory solutions, denoted by the class

$$C = \{\psi_1, \psi_2, \dots, \psi_n\}. \quad (57)$$

Each such solution minimizes a hyperbolic action of the form

$$S[\psi] = \int dx dt \left( \frac{1}{2} \eta^{\mu\nu} \partial_\mu \psi^* \partial_\nu \psi - \frac{1}{2} m^2 \psi^* \psi \right). \quad (58)$$

At this stage, the action should be read as an admissibility functional selecting structurally stable oscillatory forms, rather than as a complete microscopic law of physical dynamics.

These solutions may coexist, nest within one another, or compete for admissible support. All of them correspond to admissible configurational trajectories within pseudo-Euclidean organization, representing distinct modes of compensation, symmetry realization, and dynamic stabilization.

## Admissibility functional in a pseudo-Euclidean frame

Selection among admissible configurations is governed by an admissibility functional  $\mathcal{F}[\psi]$  defined within the chosen pseudo-Euclidean frame, expressed as

$$\mathcal{F}[\psi] = \int d^2x \sqrt{-\det \eta} (\alpha_1 \chi_{stab}(\psi) + \alpha_2 \chi_{phase}(\psi) + \alpha_3 \chi_{energy}(\psi) + \alpha_4 \chi_{closure}(\psi)). \quad (59)$$

Each metric  $\chi_i$  is constructed from densities associated with components of the energy-momentum structure  $T^{\mu\nu}(\psi)$ .

These quantities function here as admissibility metrics for structural selection. They are not introduced as empirically measured observables, but as formal descriptors of the conditions under which a configuration can remain admissible within the present framework.

The coefficients  $\alpha_i$  encode environmental and configurational conditions. The resulting scalar quantity is therefore not treated as a fundamental physical invariant, but as a frame-stable admissibility measure for structural selection within the present formalism. The structurally selected configuration is then defined by

$$\Psi_{selected} = \arg \min_{\psi \in C} \mathcal{F}[\psi]. \quad (60)$$

## Frame-stable ontological selection

Within the present formulation,  $\Psi_{selected}$  denotes the configuration that minimizes  $\mathcal{F}[\psi]$  in the chosen structural frame. Ontological selection is therefore treated here as frame-stable under the admissibility assumptions of the current formalism, while a fully Lorentz-covariant derivation is deferred to later development.

Selection at this stage identifies the configuration that remains most admissible under the simultaneous satisfaction of multiple structural criteria.

The mechanism of arising therefore takes the form of a functional filter operating within the field of admissibility.

## Completion of Level 0

Phase D completes Level 0 by establishing the ensemble as the first collective quantum of structural organization. What arises at this point is no longer a fluctuation, an oscillatory mode, or an isolated node, but an indivisible collective configuration endowed with its own stability, selection rules, and interaction capacity.

This transition marks the completion of structural formation at Level 0 and opens the threshold toward structural quantization. Level 1 begins when ensemble-quanta interact as newly constituted basic units, thereby establishing the ontological preconditions for quantum-field organization and for field-level dynamics. The decisive shift is that the relevant carriers are no longer primitive oscillatory closures internal to Level 0, but collectively stabilized units whose interactions are now admissible in effective field form.

## Transition to Level 1: ontological foundation of quantum-field organization

At the next ontological level, ensemble-quanta begin to interact with one another as newly constituted basic units. To describe these interactions, an effective field  $\Phi(X, t)$  is introduced, where the abstract coordinate  $X$  labels the position of an ensemble-quantum within the configurational domain.

An effective admissibility Lagrangian at the next level takes the form

$$\mathcal{L}_{eff} = \frac{1}{2}(\partial\Phi)^2 - V(\Phi), \quad (61)$$

where the potential  $V(\Phi)$  encodes the admissible contributions of different ensemble-quantum types and their collective interaction patterns. This establishes the hierarchy

$$\phi_p \rightarrow \text{ensemble-quantum} \rightarrow \Phi \quad (62)$$

as the ontological foundation of the first quantum-field level.

## Conclusion

Level 0 has been reconstructed as an ontological sequence in which distinguishability, compensation, oscillatory stability, resonant closure, node formation, and collective integration become admissible in successive phases. The analysis begins not from presupposed spacetime, particles, or established physical law, but from the minimal structural conditions under which difference can be preserved, stabilized, and organized into bounded configurational units. Within this reconstruction, time and space first appear as directional conditions of distinguishability, energy as a compensatory measure of admissible deviation, mass as an integrability condition of oscillatory closure, and the resonant node as the first localized unit of structurally persistent form.

The completion of Level 0 occurs when isolated resonant nodes become integrated into ensemble-quanta, that is, into collective configurational units whose stability cannot be reduced to the properties of single nodes taken independently. At this point, arising acquires bounded collective organization and becomes capable of supporting a new ontological layer in which such ensemble-quanta function as basic carriers. What is completed here is therefore not yet a theory of quantum fields, but the ontological groundwork required for field-level organization to become admissible.

The principal result of the present formalism is a reconstruction of the structural preconditions under which physical organization becomes possible. The mathematical expressions introduced throughout the paper function as admissibility formalisms and structural selection devices: they compactly encode the conditions under which configurations persist, close, and become ontologically selectable.

On that basis, Level 1 begins not from primitive oscillatory closure itself, but from collectively stabilized ensemble-quanta whose interactions become admissible in effective field form. The transition is therefore not a direct continuation of wave dynamics, but the opening of a new carrier domain built upon the completed collective organization of Level 0.